Emerging Market Bond

Delta VIX, Oil Returns, and Delta Credit Spreads-Driven Trading Strategy

Can changes in VIX, credit spreads, and oil predict next-day EM bond returns, and can those predictions generate a profitable trading strategy?

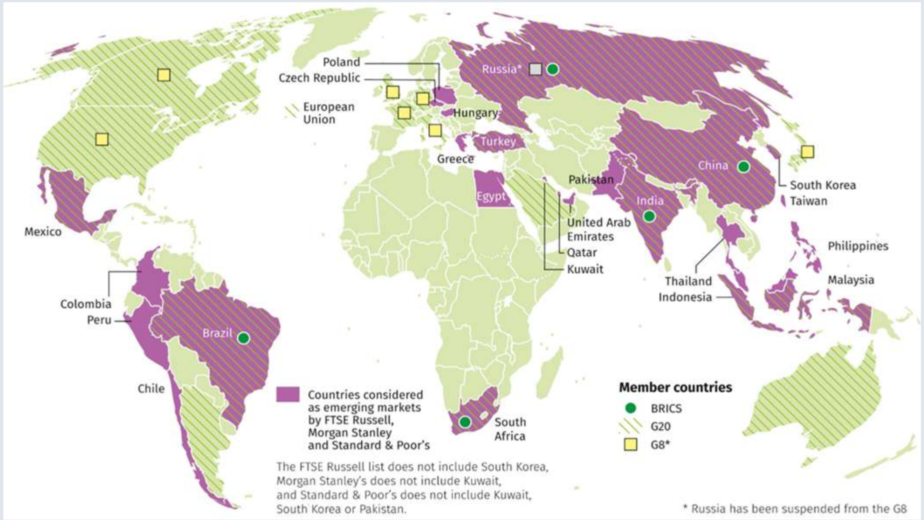
OVERVIEW

What Are Emerging Market Bonds

01 Bonds issued by less stable but developing countries like Turkey, Brazil, and Indonesia

02 Riskier than US Treasury bonds due to a variety of factors (political/economic/social)

03 We study the iShares J.P. Morgan USD Emerging Markets Bond ETF (EMB). EMB is the ETF; it tracks an index of USD-denominated EM bonds.

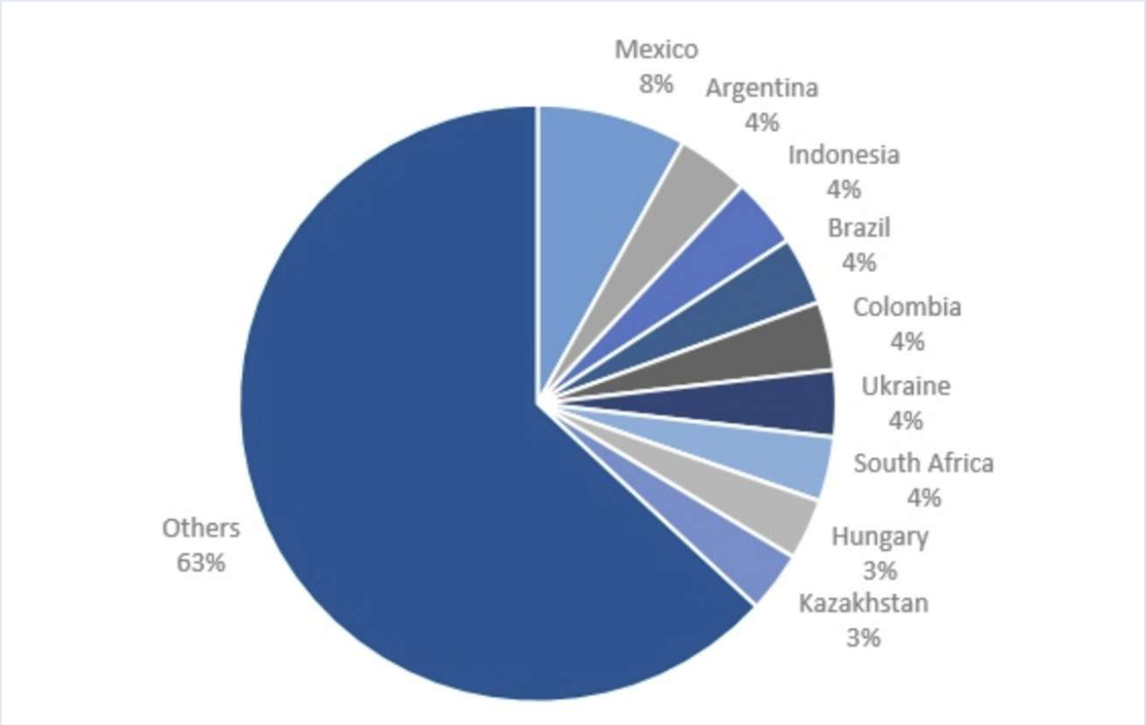


OVERVIEW

Composition: JPMorgan EMBI GD Index

The index provides exposure to liquid, US dollar-denominated emerging market sovereign debt.

The following graph shows some, but not all of the countries' bonds that make up its portfolio.



Our Research Question

Can we predict when EM bonds recover after a flight-to-safety panic, and how can we profit from it?

We Tested 3 Fear Signals

Delta (Change in) VIX
Cboe Volatility Index tracks forward expectations of volatility in the US market

Delta Credit Spreads
The credit spread index we used tracks the difference between BB U.S. high-yield credit spread over Treasuries.

Oil Returns
Crude oil futures track changes in the price of oil

We Tested 3 Models

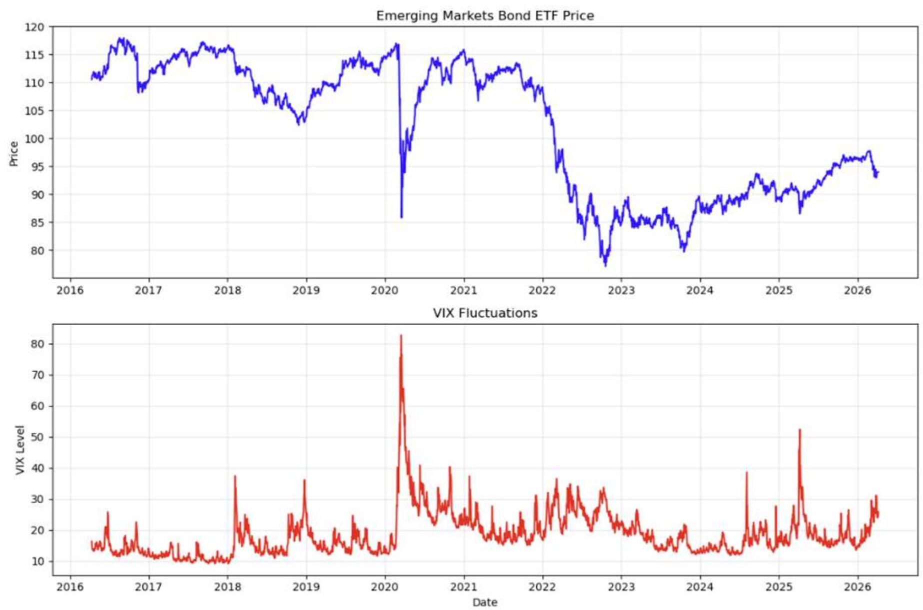
First Model - Delta VIX Only
We used a one-factor model with only continuous changes in VIX to predict EMB returns

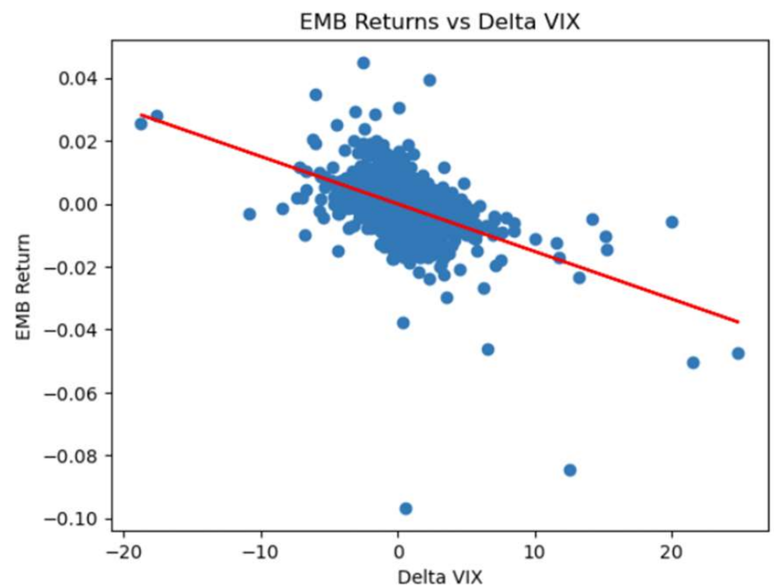
Second Model - All 3 Signals
We used a three-factor model continuously trading on changes in oil prices, VIX, and credit spreads

Model 1
Single Factor OLS

Model 1 - INTRO

Why Use VIX As Our Independent Variable





MODEL 1 - INTRO

Model 1: Setup & Results

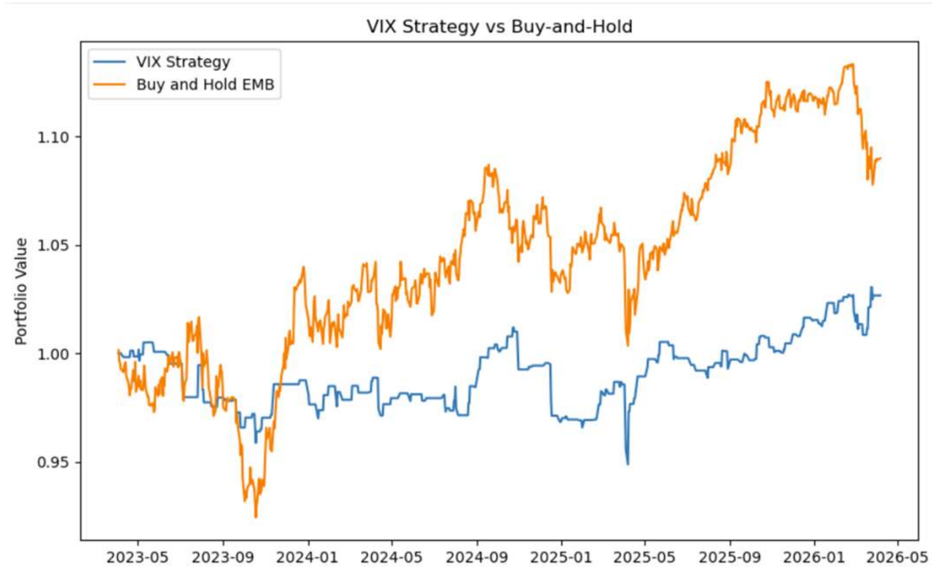
$$\hat{y} = -0.0001 + 0.0003 \cdot \Delta VIX(t) + \varepsilon(t)$$

Methodology

- Bivariate OLS regression: Δ VIX (today) → EMB return (tomorrow)
- 70% train / 30% test on daily data 2016–2026
- Predicts t+1 returns — eliminates look-ahead bias
- Strongest single predictor: ΔVIX (corr = −0.47 same day)
- On the right, we tested this strategy versus a traditional buy-and-hold

Key Insight — Sign Reversal

Correlation is negative (−0.47) but regression coefficient is positive (+0.0003). Why? Correlation is same-day panic. Regression predicts next day — when fear subsides, prices bounce back.



MODEL 1 - RESULTS

Model 1 (continued)

+0.0003
Δ VIX Coefficient

0.181
P-Value

-1.01%
Test R-Squared

0.21
Annualized Sharpe Ratio

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                 target   R-squared:                       0.006
Model:                            OLS   Adj. R-squared:                  0.005
Method:                 Least Squares   F-statistic:                     1.789
Date:                Tue, 21 Apr 2026   Prob (F-statistic):              0.181
Time:                        20:15:10   Log-Likelihood:                 6255.2
No. Observations:                1757   AIC:                        -1.251e+04
Df Residuals:                    1755   BIC:                        -1.250e+04
Df Model:                           1
Covariance Type:                  HC3
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.0001      0.000     -0.860      0.390      -0.000       0.000
delta_vix      0.0003      0.000      1.338      0.181      -0.000       0.001
==============================================================================
Omnibus:                     1304.933   Durbin-Watson:                   1.894
Prob(Omnibus):                  0.000   Jarque-Bera (JB):           119776.293
Skew:                          -2.748   Prob(JB):                         0.00
Kurtosis:                      43.074   Cond. No.                         2.04
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```

Negative Test R^2 (-1.01%) is standard in financial return prediction because of high noise in daily EMB returns. ΔVIX had the strongest same-day relationship with EMB and a positive next-day coefficient (thereby suggesting possible short-term rebound after volatility shocks), but the coefficient was not statistically significant after applying HC3 robust standard errors.

Model 2
All Three Signals

MODEL 2 - SELECTION

Why Also Use Oil or Δ Credit Spreads?

Δ Credit Spreads

Measures default probability and US investment climate.

- **Widening Spreads:** Indicates increasing risk and worsening financial conditions.
- **Return Impact:** Clear link between positive change in spreads (widening) and a **reduction in EMB returns**.

Oil Returns

Major commodity export for EMB nations.

- **Return Impact:** Sudden spikes in oil may signify massive **geopolitical volatility**.
- **One Complication:** Slow longer-term rises in the price may signify a stronger economy and currency for exporting countries.

By focusing on the delta (change) of these variables, we capture the momentum and shock factors essential for predicting short-term emerging market bond performance.

DATA

Dataset & Feature Engineering

2016-2026	2,500+	3	70 / 30
Date Range	Daily Observations	Predictors	Train / Test Split

Variable	Constructed As	Role	Intuition
em_return (Y)	log(EMBt / EMBt-1)	Dependent variable	What we predict and trade
Δ VIX (X_1)	$VIX_t - VIX_{t-1}$	Primary predictor	Δ in levels; fear spike detector
Δ Credit Spread (X_2)	Spreadt − Spreadt-1	Risk premium proxy	Δ in levels; increased risk off climate
Oil Return (X_3)	log(OIL_t / OIL_t-1)	Global risk appetite	Δ in levels; increased macro uncertainty

Returns used for prices (oil, EMB); first differences used for non-prices (VIX, spreads). Eliminates look-ahead bias by predicting t+1 returns using t predictors.

DATA - CORRELATIONS

How the Variables Move Together

Δ VIX vs EMB Return	**−0.47 (Strong Negative)**	On the SAME day: when fear spikes, investors panic-sell EMBs. This is a contemporaneous reaction, not a prediction.
Δ Credit Spread vs EMB	**−0.35 (Moderate Negative)**	Higher credit spreads = more market stress = investors dump risky EM bonds on the same day they see spreads widen.
Oil Return vs EMB Return	**+0.24 (Weak Positive)**	Rising oil = healthy global economy = more risk appetite. But the relationship is noisy since geopolitical volatility increases prices.
Δ VIX vs Δ Credit Spread	**+0.46 (⚠ Multicollinearity)**	PROBLEM: Two of our predictors are highly correlated with each other; they explain the same variation in EMB returns.

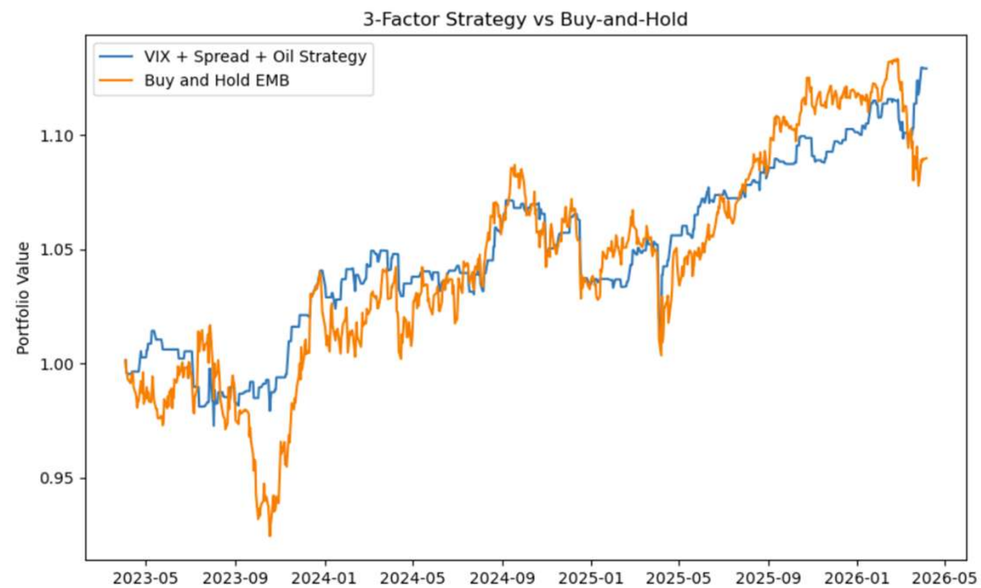
Important: correlations are same-day (time t) relationships which is good for understanding reactions, but NOT as good for predicting future returns.

MODEL 2 - INTRO

Model 2: Setup & Results

Methodology

- Using 3 factors to predict EMB Returns, and we have 2 choices: long (only invest today if predicting EMB return tomorrow is positive) or stay out (do nothing)
- Corrected for heteroskedasticity by using HC3 (Heteroskedacity-Consistent standard errors type 3) - helps provide more reliable p-values by ensuring outliers don't make your p-values look smaller
- The test R-Squared decreased from -0.0101 (for delta_vix only) to -0.0162 → this means the multivariate model became less accurate in predicting EMB returns
- We plotted the cumulative returns of our strategy and compared it to a buy and hold strategy



Negative Test R^2 (-1.01%) is standard in financial return prediction because of high noise in daily EMB returns. The key finding is that Δ VIX is statistically significant and directionally correct for short-term reversals.

MODEL 2 - 3 FACTOR MODEL

Model 2 Regression Statistics

3 Factor Model - Delta Vix, Delta Credit Spreads, & Oil Returns

- Test R-Squared = -0.0162
- Annualized Sharpe Ratio = 0.84

0.0104
Oil Return Coefficient

0.0003
Delta VIX

0.0016
Delta Spread

-0.0001
Constant

```
Equation: ŷ = 0.0104 * (oil_return) + 0.0003 * (delta_vix) + 0.00160 * (delta_spread) - 0.0001
```

MODEL 2 - MULTI-FACTOR OLS

Model 2 Analysis

$$\text{EMB Return\_t+1} = \alpha + \beta_1 \cdot \Delta\text{VIX\_t} + \beta_2 \cdot \Delta\text{Spread\_t} + \beta_3 \cdot \text{Oil\_Return\_t} + \varepsilon$$

What Improved

- Used HC3 heteroskedasticity correction for more reliable p-values
- Strategy: go long tomorrow if predicted EMB return > 0, else stay out
- Annualized Sharpe ratio: ~0.20 (model 1) vs 0.84 (model 2)

Strategy vs Buy-and-Hold

- 3-factor strategy ends at 1.129 × vs buy-and-hold at 1.09×
- Sep 2023: strategy protected capital when buy-and-hold crashed
- Strategy consistently hovers at or above buy-and-hold

The Problem: Multicollinearity

- Δ VIX and Δ Credit Spread are 0.46 correlated — shared signal
- Condition number = 84 (>30 = very high multicollinearity)
- Result: none of the 3 predictors are individually significant
- More variables did not improve prediction — double-counting noise
- Fix: Ridge regression adds penalty term to shrink correlated predictors

Lesson: adding correlated predictors inflates the condition number and destroys individual significance, even when R^2 marginally improves.

Model 3
All Three Signals
with Ridge

MODEL 3 – Ridge 3 FACTOR MODEL

Model 3: Ridge 3 Factor Model

3 Factor Model (Ridge) - Delta Vix, Delta Credit Spreads, & Oil Returns

- Test R-Squared = -0.0149
- Annualized Sharpe Ratio = 0.89

0.000245
Oil Return Coefficient

0.000512
Delta VIX

0.000160
Delta Spread

-0.000141
Constant

```
Equation: ŷ = 0.00937428* (oil_return) + 0.00025108 * (delta_vix) +
0.00160883* (delta_spread) - 0.00013946
```

*Note: Coefficients in the boxes are standardized coefficients – these are NOT the original unit-coefficients (see footnote).

MODEL 3 – Ridge 3 Factor Model

Model 3 Analysis

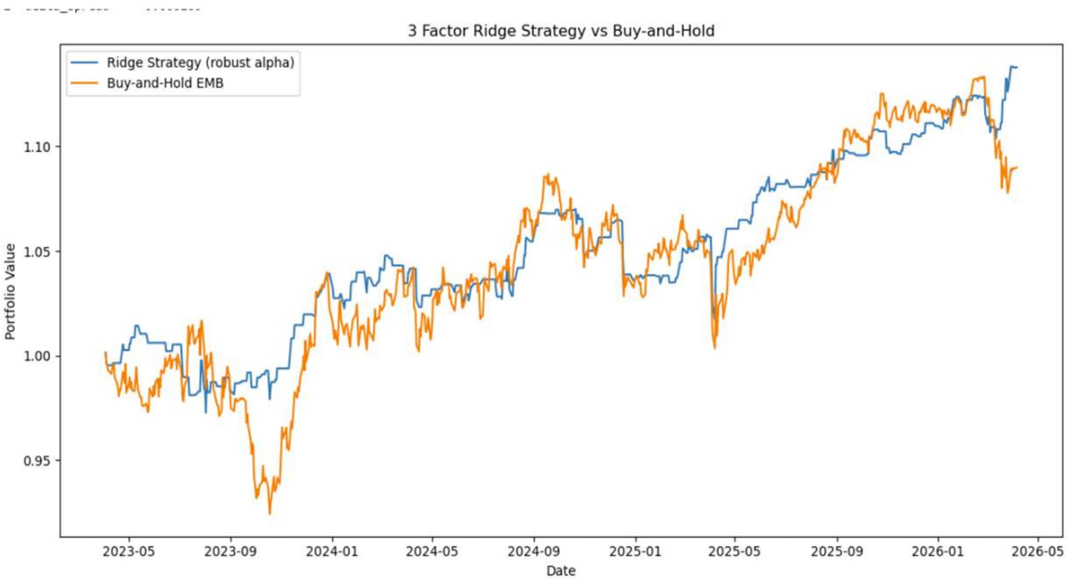
$$\text{EMB Return\_t+1} = \alpha + \beta_1 \cdot \Delta\text{VIX\_t} + \beta_2 \cdot \Delta\text{Spread\_t} + \beta_3 \cdot \text{Oil\_Return\_t} + \varepsilon$$

What Improved

- Annualized Sharpe ratio: ~0.84 (model 2) vs 0.89 (model 3)
- Returns: 12.9% (model 2) vs 13.8% (model 3)
- Penalty term to shrink correlated predictors – more stable
- Test R-Squared: -0.0162 (model 2) vs -0.0149 (model 3)

Strategy vs Buy-and-Hold

- Ridge 3-factor strategy ends at 1.138 × vs buy-and-hold at 1.09×
- Sep 2023: strategy protected capital when buy-and-hold crashed
- Strategy consistently hovers at or above buy-and-hold



Lesson: ridge regression improved returns and Sharpe ratio by adding penalty terms to shrink correlated predictors and reduce the model's sensitivity to overlapping predictor information.

MODEL 2 - RESULTS

Ending returns comparing Models 1,2, and 3

date	cum_return_vix	cum_return_full	cum_return_ridge_robust
2026-03-30	1.026676	1.129538	1.138103
2026-03-31	1.026676	1.129538	1.138103
2026-04-01	1.026676	1.129538	1.138103
2026-04-02	1.026676	1.129178	1.137740
2026-04-06	1.026676	1.129178	1.137740

date	cum_buy_hold
2026-03-30	1.089170
2026-03-31	1.088358
2026-04-01	1.089634
2026-04-02	1.089286
2026-04-06	1.089865

All strategies started off with $1. The following table shows how much we end up with at the end of the 30% testing data period.

- 1 factor model (delta_vix only) → Portfolio value ≈ 1.027 → 2.7% return
- 3 factor model (no ridge) → Portfolio value ≈ 1.129→ 12.9% return
- 3 factor model (ridge) → Portfolio value ≈ 1.138 → 13.8% return
- Buy & Hold strategy → Portfolio value ≈ 1.09 → 9.0% return

MODEL COMPARISON

Overview of the Three Models

	Model 1 OLS (1-factor)	Model 2 OLS (3-factor No Ridge)	Model 2 OLS (3-factor No Ridge)
Approach	Regression	Regression	Ridge Regression
Signal	Δ VIX	Δ VIX + Spreads + Oil	Δ VIX + Spreads + Oil
Prediction	Tomorrow's return	Tomorrow's return	Tomorrow's return
Test R^2	-1.01%	-1.62%	-1.49%
Sharpe	0.20	0.84	0.89
Return	2.7%	12.9%	13.8%
Best Feature	Stat. significance	Beats buy-and-hold + Model 1	Beats buy-and-hold + Models 1 & 2
Limitation	Single factor	Multicollinearity	

RISKS & LIMITATIONS

Key Risks in the Trade & Limitations in the Models

OLS Assumptions are Violated

Autocorrelation, heteroskedasticity, and fat tails are present. HC3 improves inference under heteroskedasticity but doesn't address nonlinear relationships or changing market regimes.

VIX is already priced in

VIX is a widely observed measure of market risk, so changes in VIX can be incorporated into asset prices very quickly. As a result, much of the contemporaneous relationship between VIX and EMB returns are already be priced in.

USD/FX Risk Omitted

A strong USD increases the local-currency burden of paying off USD-denominated debt for sovereign EMs. This model doesn't include dollar strength or country-specific currency conditions.

Trading Costs

We assumed costless entry/exit, but we need to consider bid-ask spreads and broker costs because trading is costly. Incorporating these costs could materially reduce the advantage of our strategy vs buy and hold.

Not Fully Capturing Total Returns

EMB returns are calculated from historical closing prices rather than dividend-adjusted prices. Because EMB makes monthly distributions, the strategy measures price returns rather than total investor returns. Since EMB makes cash distributions via dividends, these aren't captured in the returns we calculate.

MODEL 1 & 2 — OLS ASSUMPTIONS

OLS Assumptions Are Violated in Practice

Autocorrelation

If model underpredicts during a crash, it also underpredicts the next day. Durbin Watson statistics slightly below 2 for all regression outputs, so there is mildly positive first-order autocorrelation. (See footnote for more info)

Heteroskedasticity

Volatility clustering: tiny errors in calm markets, massive errors during crises. Error variance is NOT constant over time.

Normal Residuals

Financial data has fat tails (high kurtosis > 3). Extreme events happen far more often than a normal distribution predicts.

Multicollinearity

Δ VIX and Δ Credit Spread are 0.46 correlated. They share explanatory power, inflating standard errors in multi-factor models.

Kurtosis reference: Normal distribution has a kurtosis of 3. Financial data has kurtosis levels typically around 5–15+, meaning extreme moves are far more common than OLS assumes.

What's Next?

Potential extensions to improve predictive power

Geopolitical Risk Index

- Caldara-Iacoviello GPR Index quantifies geopolitical/geoeconomic risk by analyzing international newspaper coverage.
- Current model does not capture geopolitical risk directly.
- Test if changes in geopolitical risk provide incremental out-of-sample predictive power.

Rolling Model Re-Estimation

- The current 70/30 split assumes factor relationships remain stable over the entire period.
- Relationships among factors may change as market regimes and monetary conditions evolve.
- Re-estimate the model periodically using a rolling window to account for changing factor relationships over time.

Move to Country-Level EM Bonds

- EMB aggregates bonds across countries with very different exposures to commodities, geopolitics, currencies, and capital flows.
- Aggregation dilutes country-specific relationships between macro shocks and bond returns.
- Extend the strategy to country-specific EM bond indices and construct a portfolio based on predicted returns.

Add USD/EM-FX Factor

- A stronger U.S. dollar can increase the local-currency burden of servicing USD-denominated debt for emerging-market sovereigns.
- More pressure on sovereign bond prices due to higher capital outflows and tighter financial conditions in emerging markets.
- A USD or EM-FX factor could help capture the effect of dollar strength and EM currency flows.

Key Takeaways

Main Lessons Learned

1. Statistical Fit ≠ Trading Performance

- All models had negative test R^2, showing that next-day EMB return magnitudes are difficult to predict. However, the multifactor models generated substantially stronger risk-adjusted trading performance.

2. Combining Factors Improved the Trading Signal

- VIX alone produced limited economic value, while combining VIX, credit spreads, and oil generated a much stronger trading strategy.

3. Ridge Provided a Modest Improvement

- Ridge regularization slightly improved both out-of-sample R^2 and Sharpe relative to the unregularized 3-factor model, suggesting some benefit from coefficient shrinkage.

4. Results Are Promising, but Not Yet a Deployable Strategy

- The Ridge strategy ended the test period at approximately 1.138 versus 1.090 for buy-and-hold, but transaction costs, cash returns, and EMB distributions are not incorporated.

Thank You

Questions?

-1.49%	0.89	13.8%
Test R^2	Annualized Sharpe Ratio	Return